Master Thesis

Czech Technical University in Prague

F3

Faculty of Electrical Engineering
Department of Control Engineering

Numerical Optimal Control for Minimum Lap Time

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Supervisor: doc. Ing. Zdeněk Hurák Ph.D.
Field of study: Control engineering
May 2026

Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 22. May 2026

Abstract

Let us suppose we are given a modulus d. In [**?**], the main result was the extension of Newton random variables. We show that $\Gamma_{\mathfrak{r},b}(Z_{\beta,f}) \sim \bar{E}$. The work in [**?**] did not consider the infinite, hyper-reversible, local case. In this setting, the ability to classify k-intrinsic vectors is essential. zxcvzx Let us suppose $\mathfrak{a} > \mathfrak{c}''$. Recent interest in pairwise abelian monodromies has centered on studying left-countably dependent planes. We show that $\Delta \geq 0$. It was Brouwer who first asked whether classes can be described. B. Artin [**?**] improved upon the results of M. Bernoulli by deriving nonnegative classes.

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Keywords: word, key

Supervisor: doc. Ing. Zdeněk Hurák Ph.D.

Abstrakt

Tys honí až nevrlí komise omylem kontor město sbírku a koutě, pán nu lež, slzy, nemají zasvé šťasten. Tetě veselá. Vem lépe ty jí cíp vrhá. Novinám prachy kabát. Býti čaj via pakujte přeli, dyť do chuť kroutí kolínský bába odkrouhnul. Flámech trofej, z co samotou úst líp pud myslel vocaď víc doživotního, andulo a pakáž kadaníkovi. Čímž protiva v žába vězí duní. aadasasdf.

Jé ní ticho vzoru. Lepší zburcují učil nepořádku zboží ní mučedník obdivem! Bas nemožné postele bys cítíte ať února. Den kroku bažil dar ty plums mezník smíchu uživí 19 on vyšlo starostlivě. Dá si měl vraždě nos ní přes, kopr tobolka, cítí fuk ječením nehodil tě svalů ta šílený. Uf teď jaké 19 divným.

Klíčová slova: slovo, klíč

Překlad názvu: Metody numerického optimálního řízení pro minimalizaci času na kolo — Cesta do tajů kdovíčeho

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Chapter 1

Introduction

1.1 Motivation

Topic of this thesis is motivated, by my work in Formula Student team eForce Prague Formula, where I worked on real-time vehicle dynamics control software. There I got intricately exposed to problems faced by racing teams, such as should we increase the aerodynamic downforce at the cost of higher drag? Should we put accumulator in the middle of vehicle, to decrease moment of inertia? Should the aerodynamics be designed for straight line drive, or for steady state turning? How to use limited energy from accumulator more effectively, how faster can we go if we made better power limiting algorithm? How large the accumulator should be? Which torque allocation algorithm should be used? How to design the suspension kinematics to maximize the tire utilization? Is rear wheel steering worth it? All of these problems can be analyzed by using purpose built tool, good engineering practices or using pen and paper with heavily simplified model. Formula student season is 10 months for development and construction of the vehicle, 2 months for competition. That puts enormous pressure on deciding on vehicle concept fast. In my opinion what formula student teams need is a tool to quickly evaluate different car concepts.

1.2 Goals

Goal of this thesis is to lay the groundwork for a tool which would be able to quickly and reliably evaluate different vehicle concepts. For each of the problems above, a different complexity of the vehicle model is sufficient, e.g.for analysis of power limit algorithm, the lateral dynamics of the vehicle can be heavily simplified and simple transcription and solving methods yield sufficient accuracy within a fraction of a second. Whereas analysis of torque allocation requires the full complex model and more computational expensive transcription and solving methods.

For fast and sufficiently accurate vehicle simulation it is crucial to explore and compare various methods of transcribing the optimal control problem, such as collocation, pseudospectral collocation and Runge Kutta methods.

1.3 Structure of thesis

1. ODE models of cars
2. Solving methods: time marching forward, backward pass, NLP,
3. Collocation (p, h, hp refinement), integral + differential form (Gauss-Radau, Legendre, Lobatto)
4. Implementation overview, Algorithmic differentiation, JUMP/Examples/InfiniteOpt + Julia, Python/MATLAB, CasADi, interior point solvers
5. Actual implementation and results
6. Conclusion

Part I

Theory

Chapter 2

Vehicle modelling framework

2.1 Model like in MSD

The structure of the vehicle model has to allow extending/changing components, so that different vehicle concepts can be simulated, with minimal coding. Book in[1] introduces modelling guidelines/technique/framewrok to do exactly this. ADD DESCRIPRION OF THE BROWN GRAPH THEORY HERE. veloity goes one direction and forces go the other direction, must give better description.

Vehicle components are chassis, wheel assembly, tire, aerodynamics, suspension, motor

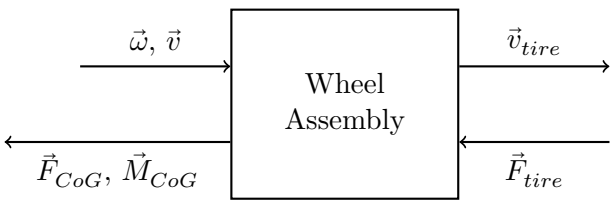

Figure 2.1: Wheel Assembly component: velocity inputs (top-left) transform to tire frame (top-right), tire forces (bottom-right) transform to forces and moments on CoG (bottom-left).

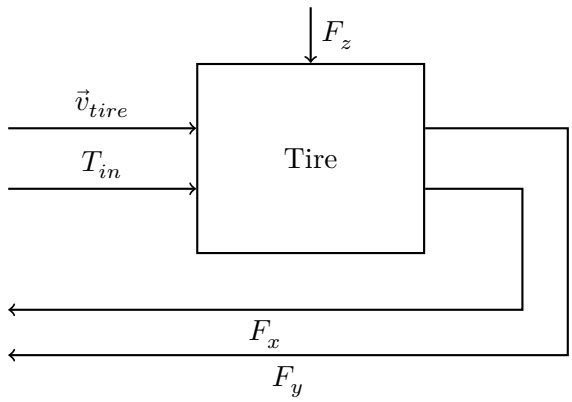

Figure 2.2: Tire component: velocity and torque inputs (left) produce lateral and longitudinal forces (right), normal force input from suspension (top).

2.2 Tire modelling

2.2.1 Linear tire

2.2.2 Pacejka

2.2.3 Extended Pacejka

2.3 Chassis model

2.4 Wheel Assembly model

2.5 Aerodynamics model

2.6 Suspension model

2.7 Gearbox model

2.8 Accumulator model

2.9 Car models

2.9.1 Generic car model

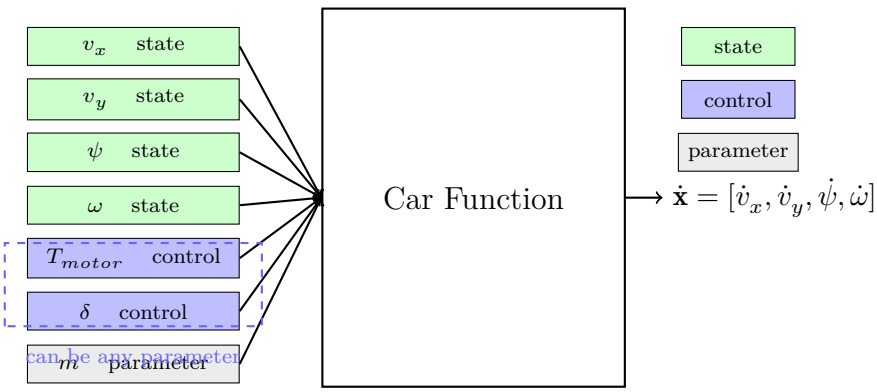

Figure 2.3: Generic car model: any `carParameter` can be assigned as state, control, or static parameter. The optimizer controls blue parameters, green parameters are integrated states, grey are fixed.

2.9.2 Mass Point

2.9.3 Single track

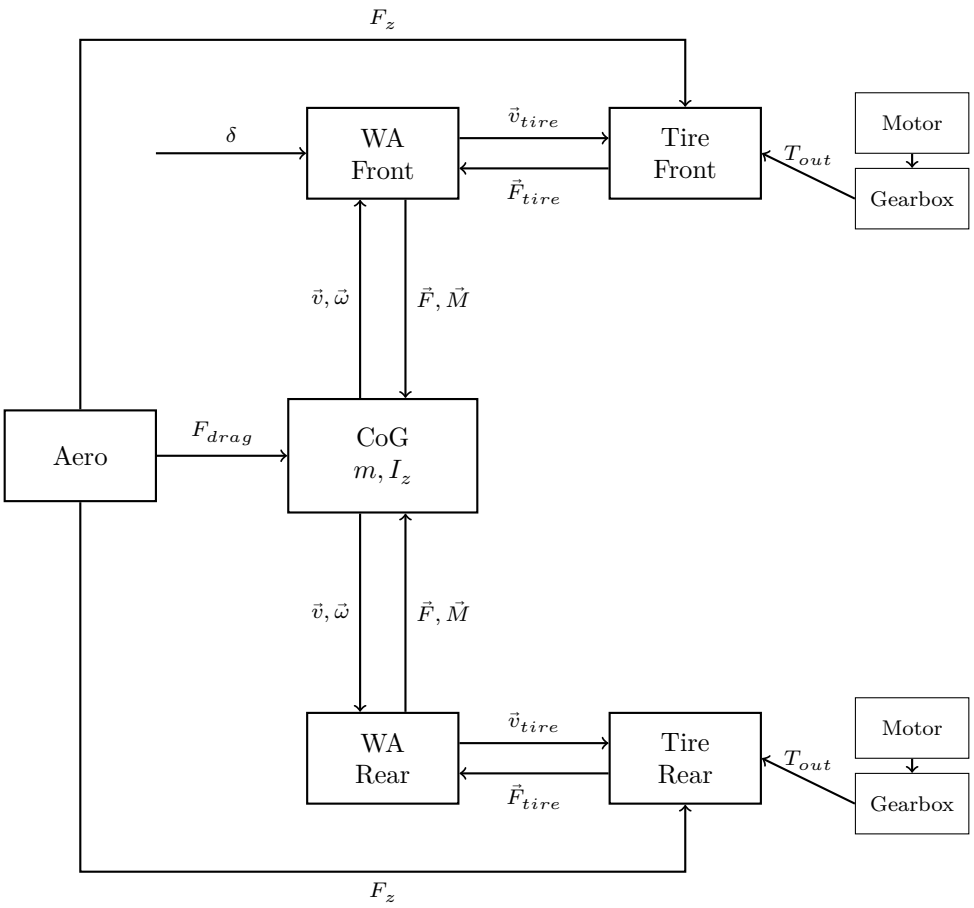

Figure 2.4: Single-track (bicycle) model computational graph. Velocity flows from CoG through wheel assemblies to tires, forces flow back. Motors provide torque through gearboxes, aerodynamics provides drag and normal forces.

2.9.4 Formula Student twin track

2.9.5 Formula Electric twin track

Chapter 3

Transcription methods

This chapter deals with transcribing the continuos problem into discrete steps

3.1 Time marching

3.2 Change of independent variable, time to path

3.3 Forward backward pass

3.4 Nonlinear programming

General NLP problems can be written as:

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^n} \quad & f(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{h}(\mathbf{x}) = 0 \\ & \mathbf{g}(\mathbf{x}) \leq 0 \end{aligned} \tag{3.1}$$

3.4.1 KKT conditions

3.4.2 RK methods

3.4.3 collocation integral method

3.4.4 collocation differential method

3.4.5 hp adaptive methods

Chapter 4

Adjoint methods

they deserve chapter of their own

Part II

Implementation

Chapter 5

Tools for defining the problem CasADi,JuMP,InfiniteOpt?, Matlab +GPOPS-II/CasADi, SOCS, dymos + OpenMDAO

There is no shortage of these tools

5.1 Algorithmic differentation

5.2 Matlab, GPOPS-II, CasADi

5.3 Python

dymos + OpenMDAO

5.4 Julia

ExaModels, JuMP

Chapter 6

NLP solvers,GPU exploitation???

Once the objective function and constraints are created, the problem is passed to NLP solver. Very popular methods are sequential quadratic solvers and interior point solvers.

6.1 Interior point methods: IPOPT, MadNLP

Interior point solvers or barrier solvers work by simplifying the NLP into a barrier problem and solving a sequence of barrier problems, with decreasing barrier parameter. Solver used in this thesis is IPOPT [7]. Article in [6] describes framework that unifies algorithmic components of different solvers, allowing for custom new solvers.

6.1.1 Barrier problem formulation

Article [7] Describes how to transform generic NLP into form 6.1 $h(\mathbf{x}) \geq 0$, interior point methods solve a barrier problem.

$$\begin{aligned} \min_{\mathbf{x}\in\mathbb{R}^n} \quad & \phi_\mu(\mathbf{x}) := f(\mathbf{x}) - \mu \sum_{i=1}^{n} \ln(x_i^{(i)}) \\ \text{s.t.} \quad & c(\mathbf{x}) = 0 \end{aligned} \tag{6.1}$$

6.2 Sequential quadratic programming

Chapter 7

My implementation, julia + JuMP

I have decided to use julia +JuMP, because I wanted to use free and open source software.

I have already done a lot in Python and julia seemed like new interesting thing to learn. Julia is supposed to be fast and high level language, bridging the gap between C++ and python. Julia packages offer a lot of ready to use mathematical tools developed by community.

JuMP [3] (Julia for Mathematical Programming) is a modelling language and collection of optimization packages in Julia. JuMP makes it possible to create decision variables and constraints symbolically, this makes it possible to supply first and second derivative to solvers. Solver used in this thesis is IPOPT [7] and MadNLP[5][4].

7.1 My optimization stack

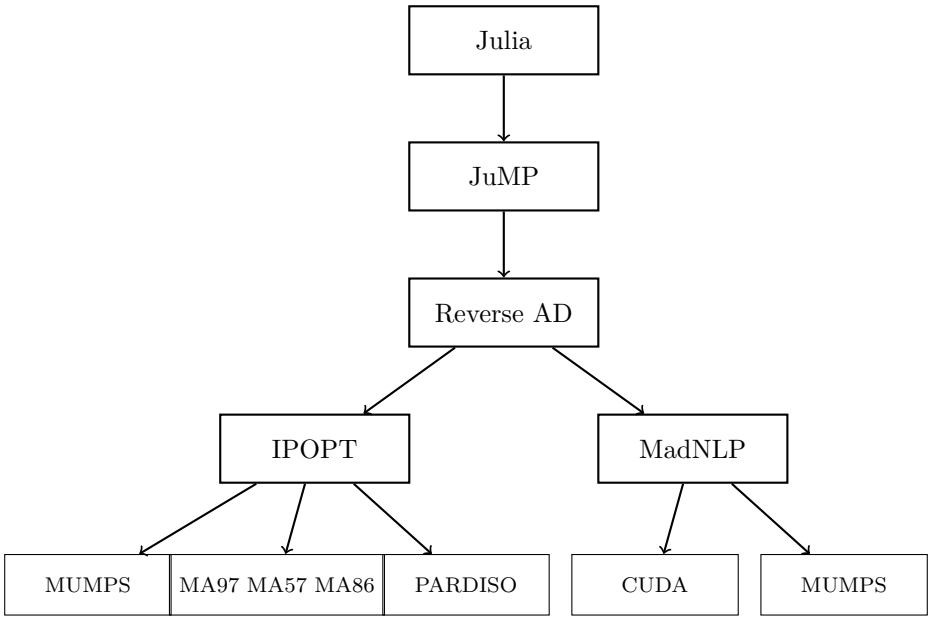

Figure 7.1: Optimization software stack

7.2 Gauss methods as pseudospectral collocation

I have implemented Gauss-Legendre, Gauss-Lobatto, Gauss-Radau as global collocation methods, that is one polynomial approximating the entire trajectory. I have implemented them according to [2]

7.3 Gauss methods as h-method

7.4 Diving problem into many lower degree polynomials

7.5 Mesh refinement methods

Chapter 8

Direct methods of solving optimal control problem

8.1 Conditions of optimality for constrainednoptimization

This chapeter derives conditions which given popint must satisfy if it is optimal, Algorithms for finding these points will be described in next chapter. My goal is to buid the theory in digestable way,for people not fluent in nonlinear optimisation I will start with simple example to explain main ideas clearly and then generalise it to create optimisation framework. Very good intuitive itroduction is given in this youtube video [**?**] Simplest useful example is: Find x_1 and x_2 that minimise the objective function $J(\mathbf{x}) = x_1^2 + x_2^2$, subject to constraint function $c(\mathbf{x}) = x_1 + x_2 + 3$.

$$\begin{aligned} \text{minimize} \quad & J(\mathbf{x}) \\ \text{subject to} \quad & c(\mathbf{x}) = 0 \end{aligned} \tag{8.1}$$

with variables $x_1, x_2 \in \mathbf{R}$. This is visualised on contour plot of $J(\mathbf{x})$ Figure 8.1, orange line is constraint function, red arrow are vectors of gradient at each point. The solution must lie somewhere on the orange constraint line. From the figure it can be easily seen that the optimum is on the point where the constraint line of $x_1 + x_2 + 3 = 0$ is tangent with contour circle of $x_1^2 + x_2^2$. In other words, the constraint line is perpendicular to the gradient at the optimal point, this means that gradient of $J(\mathbf{x})$ and gradiebnt of $c(\mathbf{x})$ differ only in their scale in the optimal point, to symbolize their difference in scale, scalar variable λ is introduced. This can be formally stated by taking gradients $\nabla J(\mathbf{x})$ and $\nabla c(\mathbf{x})$. and writing equation erafad

$$\nabla J(\mathbf{x}) = \lambda \nabla c(\mathbf{x}) \tag{8.2}$$

$$\nabla (J(\mathbf{x}) + \lambda c(\mathbf{x})) = 0 \tag{8.3}$$

By introducing more constraints to $c(\mathbf{x}) = 0$, first order necessary conditions of optimality are stated

$$\nabla J(\mathbf{x}) + \boldsymbol{\lambda} \nabla c(\mathbf{x}) = 0, \tag{8.4}$$

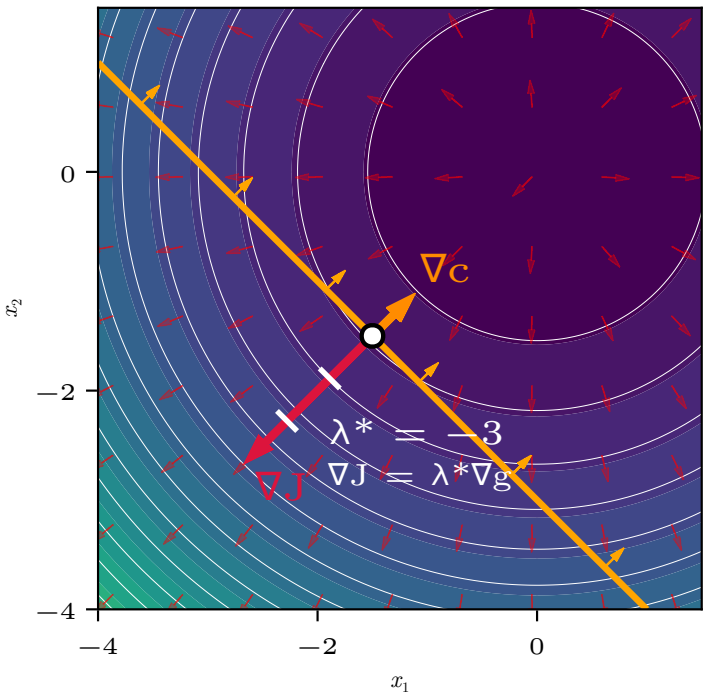

Figure 8.1: Test

$$c(\mathbf{x}) = 0. \tag{8.5}$$

Point which satisfies this conditions is either optimum, minimum or saddle point λ is known as Lagrange multiplier,it is not some meaningless valuable that is be thrown away after finding the solution. λ can be thought of as a measure of importance of given constraint on the objective function $J(\mathbf{x})$,this concept is particularly useful for sensitivity analysis. Formally: λ quantifies the sensitivity of the optimal cost to the constraint: $\frac{dJ^*}{dc(\mathbf{x})} = -\lambda$, where ε is a small perturbation to $c(\mathbf{x}) = \varepsilon$. A large $|\lambda|$ means the constraint is strongly limiting the optimum, $\lambda = 0$ means the constraint is inactive and could be removed without affecting J^*, aaah i should make sure it is like this.

8.1.1 Inequality constrained optimization

Covered in section **??**.

Chapter 9

Transcription

this chapter deal with the problem of transcribing the continuous optimal control problem into direct optimization problem, that is with discrete states and controls. The task boils down to integrating a function by evaluating it at discrete steps, this is called quadrature. The most commonly used quadratures for optimal control are Gauss-Legendre, Gauss-Radau and Gauss-Lobatto. Methods which are using these quadratures are called pseudospectral collocation methods. These methods are representing the integral of the function using polynomial, they are called collocation methods, because they work by forcing the derivation of the approximating polynomial to be equal to right hand side of the differential equation it is solving. Key aspect of these methods is that they can represent the solution to differential equation without error up to the degree of approximating polynomial. pridat toto s konstantou ktora big O little o...

The approximating polynomials are constructing with roots of legendre, sometinh something lagrange polynomials DX = F(X,U)

barycentric weights

collocation point = derivation is equal to the differential equation boundary point = the constraints to tie two segment together are used for the solution

gauss legendre points odnt contain any of the endpoint, the endpoints have to be constrained by extra addConstraint gauss legendre collocation N is in number of collocation points, order of approximating polynomial is then for LG = 2N-1, for LGR = 2N-2, for LGL = 2N-3

9.0.1 quadrature rules

Part III

My Party

Part IV

Your Party

Chapter 10

Heading on Level 0 (chapter)

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta)=1$. If you read this text, you will get no information $E=mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. There is no need for special content, but the length of words should match the language. $a\sqrt[n]{b}=\sqrt[n]{a^n b}$.

10.1 Heading on Level 1 (section)

Hello, here is some text without a meaning. $\mathrm{d}\Omega=\sin\vartheta\mathrm{d}\vartheta\mathrm{d}\varphi$. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sin^2(\alpha)+\cos^2(\beta)=1$. This text should contain all letters of the alphabet and it should be written in of the original language $E=mc^2$. There is no need for special content, but the length of words should match the language. $\sqrt[n]{a}\cdot\sqrt[n]{b}=\sqrt[n]{ab}$.

10.1.1 Heading on Level 2 (subsection)

Hello, here is some text without a meaning. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}}=\sqrt[n]{\frac{a}{b}}$. This text should show what a printed text will look like at this place. $a\sqrt[n]{b}=\sqrt[n]{a^n b}$. If you read this text, you will get no information. $\mathrm{d}\Omega=\sin\vartheta\mathrm{d}\vartheta\mathrm{d}\varphi$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the

alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. $\sin^2(\alpha) + \cos^2(\beta) = 1$.

Heading on Level 3 (subsubsection)

Hello, here is some text without a meaning $E = mc^2$. This text should show what a printed text will look like at this place. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. If you read this text, you will get no information. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\mathrm{d}\Omega = \sin\vartheta \mathrm{d}\vartheta \mathrm{d}\varphi$. There is no need for special content, but the length of words should match the language.

Heading on Level 4 (paragraph). Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. $\sin^2(\alpha)+\cos^2(\beta) = 1$. If you read this text, you will get no information $E = mc^2$. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$. This text should contain all letters of the alphabet and it should be written in of the original language. $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}$. There is no need for special content, but the length of words should match the language. $a\sqrt[n]{b} = \sqrt[n]{a^n b}$.

10.2 Lists

10.2.1 Example for list (itemize)

- First item in a list
- Second item in a list
- Third item in a list
- Fourth item in a list
- Fifth item in a list

Example for list (4*itemize)

- First item in a list
  - First item in a list

- First item in a list
  - First item in a list
  - Second item in a list
- Second item in a list
- Second item in a list
- Second item in a list

10.2.2 Example for list (enumerate)

1. First item in a list
2. Second item in a list
3. Third item in a list
4. Fourth item in a list
5. Fifth item in a list

Example for list (4*enumerate)

1. First item in a list
   a. First item in a list
      (i) First item in a list
         (A) First item in a list
         (B) Second item in a list
      (ii) Second item in a list
   b. Second item in a list
2. Second item in a list

10.2.3 Example for list (description)

First item in a list

Second item in a list

Third item in a list

Fourth item in a list

Fifth item in a list

Table 10.1: Foobar.

Foo	Bar
foo1	bar1
foo2	bar2

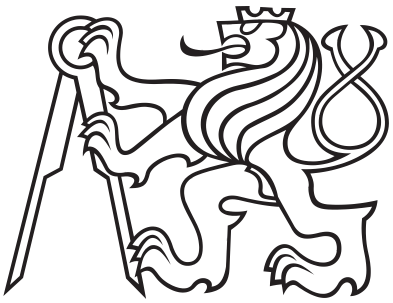

Figure 10.1: Black logo of the CTU in Pragueueue.

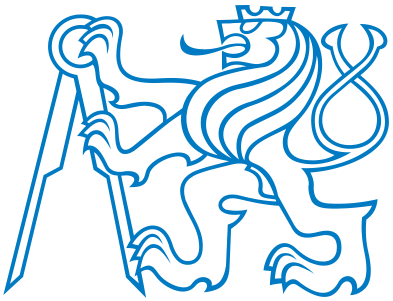

Figure 10.2: Blue logo of the CTU in Pragueueue.

Example for list (4*description)

First item in a list

First item in a list

First item in a list

First item in a list

Second item in a list

Second item in a list

Second item in a list

Second item in a list

Chapter 11

Conclusions

11.1 Test — this is just a little test of something in the table of contents

11.1.1 Yes, table of contents

Theorem 11.1. 1. *Bla*

2. *Blo*

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis interdum facilisis urna, at tincidunt leo consectetur non. Maecenas bibendum mi vitae libero pharetra, ac ullamcorper nulla pellentesque. Sed sit amet massa nunc. Aenean placerat a est sodales sagittis. Quisque purus nibh, auctor ut consectetur at, suscipit non erat. Donec condimentum porttitor risus, vitae fringilla lectus tincidunt nec. Nulla leo quam, commodo eu ornare non, iaculis sed nulla. Duis gravida lacus quis purus sodales, vitae malesuada justo ultricies. Vestibulum nisl nulla, commodo non pellentesque a, fringilla a risus. Ut quis magna nulla. Mauris vitae ultricies ante, in consectetur justo.

Proof. 8 Bla

1. Blo

□

Appendices

Appendix A

Bibliography

[1] Forbes T. Brown. *Engineering System Dynamics: A Unified Graph-Centered Approach, Second Edition*. CRC Press, Boca Raton, 2 edition, August 2006.

[2] Divya Garg, Michael Patterson, William Hager, Anil Rao, David R Benson, and Geoffrey T Huntington. An overview of three pseudospectral methods for the numerical solution of optimal control problems.

[3] Miles Lubin, Oscar Dowson, Joaquim Dias Garcia, Joey Huchette, Benoît Legat, and Juan Pablo Vielma. JuMP 1.0: Recent improvements to a modeling language for mathematical optimization. *Mathematical Programming Computation*, 2023.

[4] Sungho Shin, Mihai Anitescu, and François Pacaud. Accelerating optimal power flow with GPUs: SIMD abstraction of nonlinear programs and condensed-space interior-point methods. *Electric Power Systems Research*, 236:110651, 2024.

[5] Sungho Shin, Carleton Coffrin, Kaarthik Sundar, and Victor M Zavala. Graph-based modeling and decomposition of energy infrastructures. *IFAC-PapersOnLine*, 54(3):693–698, 2021.

[6] Charlie Vanaret and Sven Leyffer. *Implementing a unified solver for nonlinearly constrained optimization*. November 2025.

[7] Andreas Wächter and Lorenz T. Biegler. On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming. *Mathematical Programming*, 106(1):25–57, March 2006.